

Recap Checkpoint 2 — after 1.3 (Exchanging Data) — cumulative 1.1 + 1.2 + 1.3

Name: _____ Date: _ **Mark:** _____ / 30

OCR H446 cumulative recap — mixes every topic taught so far. Show all working.

Questions

Q1. State the function of the **MAR** and the **MDR** during the fetch phase of the FDE cycle. **[2]** (AO1) — 1.1.1

Q2. A laptop uses **L1, L2 and L3 cache**. Explain how cache improves CPU performance, referring to **one** difference between L1 and L3 cache. **[3]** (AO1) — 1.1.1

Q3. Explain how the operating system uses **interrupts and the interrupt service routine (ISR)** to respond to a key press while a program is running. **[3]** (AO1) — 1.2.1

Q4. A program is distributed as **bytecode** to be run by a **virtual machine**. (a) Explain **one** advantage of distributing bytecode rather than machine code. **[2]** (AO2)

(b) State **one** disadvantage. **[1]** (AO2)

[3 total] — 1.2.2

Q5. Explain what is meant by **hashing**, and give **two** reasons why hashing is used to store passwords rather than storing the passwords directly. **[3]** (AO1) — 1.3.1

Q6. A row of pixels in a monochrome image is shown below, where each letter is a pixel colour:

RRRRRGGGGGGRRRR

(a) Show how **run length encoding (RLE)** would encode this row. **[2]** (AO2)

(b) State **one** type of image for which RLE gives **little or no compression**, and explain why. **[1]** (AO2)

[3 total] — 1.3.1

Q7. A company sends confidential data over the internet to a new client it has never contacted before. (a) State the difference between **symmetric** and **asymmetric** encryption. **[2]** (AO1)

(b) Explain why **asymmetric** encryption solves the problem of exchanging keys securely with a brand-new client. **[2]** (AO2)

[4 total] — 1.3.1

Q8. A flat-file table storing orders is being normalised:

```
Order(OrderID, CustomerID, CustomerName, CustomerEmail, ProductID, ProductName)
```

Here `CustomerName` and `CustomerEmail` are determined by `CustomerID`, and `ProductName` is determined by `ProductID`. (a) State the rule a table must satisfy to be in **first normal form (1NF)**. [1] (AO1)

(b) Explain why this table is **not** in **third normal form (3NF)** and list the tables produced after normalising to 3NF. [3] (AO2)

[4 total] — 1.3.2

Q9. A `Booking` table has fields `BookingID`, `Surname`, `Town`, `Nights` and `Cost`. Write an SQL statement to display the `Surname` and `Cost` of all bookings where `Town` is 'York' **and** `Nights` is greater than 3, sorted by `Cost` in **descending** order. [4] (AO2) — 1.3.2

Q10. Data is sent across the internet using the **TCP/IP** protocol stack. (a) Name the **four layers** of the TCP/IP stack in order from top (application) to bottom. [2] (AO1)

(b) State the layer at which the **IP address** is added, and the layer responsible for **splitting data into packets / reassembly**. [2] (AO1)

[4 total] — 1.3.3

Q11. An online retailer's web pages load slowly and rank poorly in search results. The developers are considering moving some processing from **server-side to client-side**, and improving how pages are **indexed and ranked** by search engines. Evaluate how these two changes could help, referring to specific web technologies. **[3]** (AO3) — *synoptic 1.3.4 (+ 1.1/1.2 performance)*
